8 (a)

$$d_{avg} = \frac{53+67}{2} = 60 \text{ m}$$

$$\Delta d = \frac{67-53}{2} = 7 \text{ m}$$ **C1**

Percentage uncertainty $= \frac{7}{60} \times 100\% = 12\%$ (to 2sf) **A1**

(b)(i) The gravitational force exerted on Dimorphos by Didymos provides the centripetal force **M1**

$$\frac{GMm}{r^2} = mr\omega^2$$

$$\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2$$ **M1**

Rearranging and making T^2 the subject,

$$T^2 = \frac{4\pi^2 r^3}{GM}$$ **A0**

(b)(ii) The orbital radius decreased **B1**

The orbital period of Dimorphos decreased from 11.9 hours to 11.4 hours after DART's impact. Since T^2 is proportional to r^3, when the orbital period decreases, the radius of its orbit will also decrease **B1**

(c)(i) For the two patterns to be just distinguishable (or resolved), the central maximum of one must lie on the first minimum of the other. **B1**

(c)(ii)

$$b\sin\theta = \lambda$$

$$\sin\theta = \frac{\lambda}{b}$$

For the two asteroids to be just distinguishable, their angular separation from the perspective the telescope lens, $\theta_A = \frac{s}{r}$, equal to θ. Since θ and θ_A is small,

$$r = s\left(\frac{b}{\lambda}\right)$$

$$= 1189\left(\frac{208\times10^{-3}}{700\times10^{-9}}\right)$$ **C1**

$$= 3.53\times10^{8}\ \text{m}$$ **A1**

<u>Note:</u> λ can be any value between 400 nm to 700 nm. (visible light range)
Accepted values of r range from 3.53×10^8 m to 6.18×10^8 m.

(d)(i) The ROSA solar array delivers 120 W of power per kilogram of the solar array. **B1**

(d)(ii) Total mass of ROSA array $= 2\times16.95$

$= 33.9$ kg

$P = 33.9\times120$ **C1**

$= 4068$

$= 4070$ W (to 3sf) **A1**

(e)(i) Xenon has a relatively high atomic mass compared to Argon or Krypton. **B1**

B1

Hence when the Xenon ions are being expelled at higher momentum for the same ion speed, the thrust force generated is greater

OR

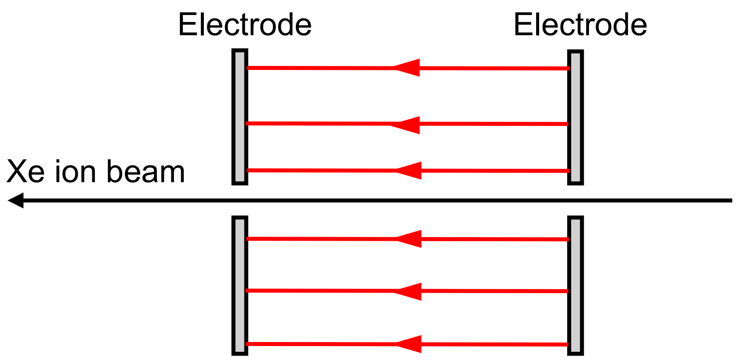
B1

Xenon has lower ionization energy than Argon or Krypton.

B1

Hence, it requires less energy to create Xenon ions to be used in the ion thruster

(e)(ii)



- 6 horizontal parallel lines, equally spaced apart pointing leftwards **B1**

(e)(iii) By Conservation of Energy,

$$\text{Gain in KE} = \text{Loss in } U_E$$
$$\frac{1}{2}mv^2 = qV$$

Since the Xe ion has 131 nucleons and has a charge of +1.60 × 10^-19 C due to having 1 electron removed,

$$\frac{1}{2}(131 \times 1.66 \times 10^{-27})(40000)^2 = \left(1.60 \times 10^{-19}\right)V$$ **C1**

$$V = 1087.3 \text{ V}$$
$$= 1100 \text{ V (to 2sf)}$$ **A1**

(e)(iv) At the maximum exhaust speed of ejection,

$$F = \frac{dp}{dt} = \frac{dm}{dt}v$$

$$\frac{dm}{dt} = \frac{F_{\text{max}}}{v_{\text{max}}} = \frac{235 \times 10^{-3}}{40000} = 5.875 \times 10^{-6}$$ **C1**

$$\text{no. of Xe ions expelled per second} = \frac{dm}{dt} \div m_{Xe}$$
$$= \frac{5.875 \times 10^{-6}}{131 \times 1.66 \times 10^{-27}}$$ **C1**
$$= 2.70 \times 10^{19}$$ **A1**